

E-Commerce Sales and Delivery Performance Analysis

Business Analytics Project Report: MySQL and Excel

Item	Detail
Data source	Brazilian E-Commerce Public Dataset by Olist (Kaggle)
Period analysed	September 2016 to October 2018
Records analysed	550,759 rows across 8 tables; 96,478 delivered orders; 93,358 customers
Tools	MySQL 8 (Workbench), Microsoft Excel

Contents

1. Company Background
2. Business Problem
3. Business Objective
4. Stakeholders
5. Business KPIs
6. Domain Knowledge
7. Dataset Overview
8. Business Questions and Findings
9. SQL Analysis
10. Dashboard Development
11. Business Recommendations
12. Business Impact
13. Conclusion and Future Scope

1. Company Background

Data owner: Olist is a Brazilian e-commerce company that connects small and medium merchants with large online marketplaces. It published a public dataset of about 100,000 anonymised orders placed between 2016 and 2018, covering customers, sellers, products, payments, deliveries and reviews. The dataset is available on Kaggle as the *Brazilian E-Commerce Public Dataset by Olist*.

Project context (simulated business scenario). The project is framed as work done by an analyst in the Commercial and Operations Analytics team of an online marketplace. Leadership wants to understand sales performance, delivery reliability, customer satisfaction and customer retention from a single, trustworthy set of numbers.

2. Business Problem

The marketplace is growing quickly, but leadership cannot yet answer three questions:

1. **Delivery reliability.** How often are orders late, where and when, and does it hurt customer satisfaction?
2. **Customer retention.** How many customers come back after their first order? Growth that depends only on new customers is expensive and fragile.
3. **Visibility.** Data is spread across nine separate tables, so there is no single view that links sales, delivery, payments, sellers and reviews.

Problem statement: *Where, when and why does the marketplace lose customer satisfaction and repeat business, and what actions would improve revenue quality and customer experience?*

3. Business Objective

Primary objective: Identify the main drivers of customer dissatisfaction and weak retention, and recommend prioritised, measurable actions.

#	Objective	Measured by
1	Measure overall sales performance and growth	Revenue, orders, AOV, monthly trend
2	Find the products and regions that drive revenue	Category and state revenue share, Pareto analysis
3	Quantify delivery performance and its link to reviews	Late %, delivery days, review score by delay
4	Identify the best and weakest sellers	Seller revenue, review score, late %
5	Measure retention and segment customers	Repeat rate, cohort retention, RFM segments
6	Deliver a one-page executive dashboard and an action plan	Excel dashboard, recommendations

4. Stakeholders

Stakeholder	Interest	What they need from this analysis
Executive leadership (CEO, COO)	Growth, profitability, brand	One-page KPI view; top risks and opportunities
Logistics and operations	Delivery speed and reliability	Late % by state and month; where to add capacity
Seller / marketplace management	Seller quality and compliance	Seller scorecards; list of weak sellers
Customer experience and support	Satisfaction, complaints	What drives low reviews; which orders to watch
Marketing and CRM	Acquisition cost, retention	Repeat rate, cohort retention, RFM segments
Category / merchandising managers	Assortment	Category revenue, concentration, review quality
Finance	Revenue quality, payment mix	Revenue trend, payment methods, freight share
Data / BI team	Reliable, reusable data	Clean data model and documented SQL

5. Business KPIs

All KPIs use **delivered orders** unless stated. “Suggested target” values are analyst proposals for discussion, not company targets.

KPI	Definition	Result	Suggested target
Total revenue	Sum of item prices (freight excluded)	R\$ 13,221,498	Track monthly growth
Delivered orders	Orders with status “delivered”	96,478	n/a
Unique customers	Distinct customer_unique_id	93,358	n/a
Average order value (AOV)	Revenue / delivered orders	R\$ 137.04	Maintain or grow
Average delivery time	Days from purchase to delivery	12.5 days	Reduce on slow routes
Estimated delivery time	Days promised at purchase	24.4 days	Calibrate by route
Late delivery rate	Delivered on a date after the estimated date	6.77%	Below 5%
Average review score	Mean of order-level review scores (1 to 5)	4.16	4.3 or higher
Repeat customer rate	Customers with 2 or more delivered orders	3.00%	5% or higher
Freight share of revenue	Freight / item revenue	16.6%	Monitor by region
Order completion rate	Share of all orders delivered	97.02%	Maintain

6. Domain Knowledge

6.1 Industry terminology

Term	Meaning
Marketplace	Online platform where many independent sellers list products
Seller / merchant	Business that sells on the marketplace and ships the product
GMV	Gross merchandise value, the total value of goods sold
AOV	Average order value
SKU	Stock keeping unit, a unique product variant
Freight	Shipping fee charged to the customer
Last-mile delivery	Final leg of delivery from the local hub to the customer
Estimated delivery date (EDD)	Delivery date promised to the customer at purchase
Boleto	Brazilian bank-slip payment, paid in cash or through online banking
Instalments (<i>parcelamento</i>)	Splitting a card payment into monthly parts, very common in Brazil
Voucher	Gift or promotional credit used as payment
Cohort	Group of customers who first purchased in the same month
RFM	Recency, Frequency, Monetary value: a customer segmentation method
Review score / CSAT	Customer satisfaction rating after delivery
Churn	A customer who stops buying

6.2 Business workflow

1. The customer browses the marketplace and places an order.
2. Payment is processed (card, boleto, voucher or debit) and the order is approved.
3. The seller is notified, then prepares and packs the items.
4. The order is handed to a logistics partner (carrier).
5. The carrier delivers to the customer and the delivery date is recorded.
6. The customer receives a satisfaction survey and leaves a review score.
7. A repeat purchase, return or complaint follows.

The dataset records each step: purchase, approval, carrier hand-over, customer delivery, estimated date and review.

6.3 Common challenges in this industry

- **Distance and infrastructure.** Long routes to the North and Northeast raise delivery time and freight cost.
- **Peak demand.** Events such as Black Friday create volume spikes that strain carriers.
- **Seller quality control.** Customers judge the marketplace, not the individual seller.
- **Low repeat purchase.** Many marketplace customers buy once, driven by price or a single need.
- **Payment friction.** Boleto must be paid before approval, which delays processing.
- **Fraud and chargebacks** on card payments.

- **Cancellations and stock-outs** when sellers cannot fulfil an order.

6.4 Industry regulations (Brazil)

This is general context, not legal advice.

Regulation	Relevance
LGPD (Law 13.709/2018)	Brazil's data protection law: customer data must be collected and used lawfully. The dataset used here is anonymised
Consumer Defence Code (CDC)	Protects consumers. Article 49 gives a 7-day right of withdrawal for distance purchases
Decree 7.962/2013 (E-commerce Decree)	Requires clear product, price and delivery information and accessible customer service
PCI DSS	Industry standard for storing and processing card data
Central Bank payment rules	Govern payment arrangements, including boleto and instant payments

6.5 Business metrics used

Revenue, AOV, order volume, delivery time, late delivery rate, review score, negative review rate, repeat rate, cohort retention, RFM segments, revenue share (Pareto) and freight share.

7. Dataset Overview

Item	Detail
Source	"Brazilian E-Commerce Public Dataset by Olist", published on Kaggle
Licence	Listed on Kaggle as CC BY-NC-SA 4.0 (non-commercial, share-alike). Check the dataset page for current terms
Time period	4 September 2016 to 17 October 2018 (order purchase dates)
Files in the source	9 CSV files. Eight are used. The geolocation file (about 1 million rows) is not used and is not included in this repository
Number of records (8 tables used)	550,759 rows. Core tables: 99,441 orders, 112,650 order items, 99,441 customer IDs
Number of features (8 tables used)	47 columns
Target variable	None. This is descriptive and diagnostic analysis. Possible targets for future modelling: late delivery (is_late) and low review score

7.1 Records, features and data types

File	Records	Features	Data types	Missing cells
olist_customers_dataset	99,441	5	2 ID, 3 text	0
olist_sellers_dataset	3,095	4	1 ID, 3 text	0
olist_products_dataset	32,951	9	1 ID, 1 category, 7 numeric	2,448
olist_orders_dataset	99,441	8	2 ID, 1 category, 5 datetime	4,908
olist_order_items_dataset	112,650	7	3 ID, 3 numeric, 1 datetime	0
olist_order_payments_dataset	103,886	5	1 ID, 1 category, 3 numeric	0
olist_order_reviews_dataset	99,224	7	2 ID, 1 numeric, 2 text, 2 datetime	145,903
product_category_name_translation	71	2	2 text	0

7.2 Missing values and how they were handled

Table / column	Missing	%	Cause	Treatment
orders.order_approved_at	160	0.16%	Orders not approved or cancelled	Loaded as NULL
orders.order_delivered_carrier_date	1,783	1.79%	Not shipped yet	Loaded as NULL
orders.order_delivered_customer_date	2,965	2.98%	Not delivered. 8 of these orders have status "delivered" (data error)	NULL. The 8 orders are excluded from delivery analysis
products.product_category_name	610	1.85%	Category not recorded	Labelled "unknown"
products name length, description length, photos	610 each	1.85%	Same 610 products	Not used
products weight and dimensions	2 each	0.01%	Not recorded	Not used
reviews.review_comment_title	87,656	88.34%	Optional free text	Removed from the cleaned file (not needed)
reviews.review_comment_message	58,247	58.70%	Optional free text	Removed from the cleaned file (not needed)

7.3 Other data quality findings

- 814 review IDs appear more than once. The pair (review_id, order_id) is unique, so reviews were averaged per order.
- 775 orders have no items, so they cannot produce revenue.
- 2,961 orders have more than one payment record and 547 orders have more than one review. Both were aggregated per order before joining.
- Two product categories had no English translation and were added manually.
- Table relationships are clean: 0 orphan product, seller or order references.
- September to December 2016 and September to October 2018 contain very few orders, so trend charts use January 2017 to August 2018.

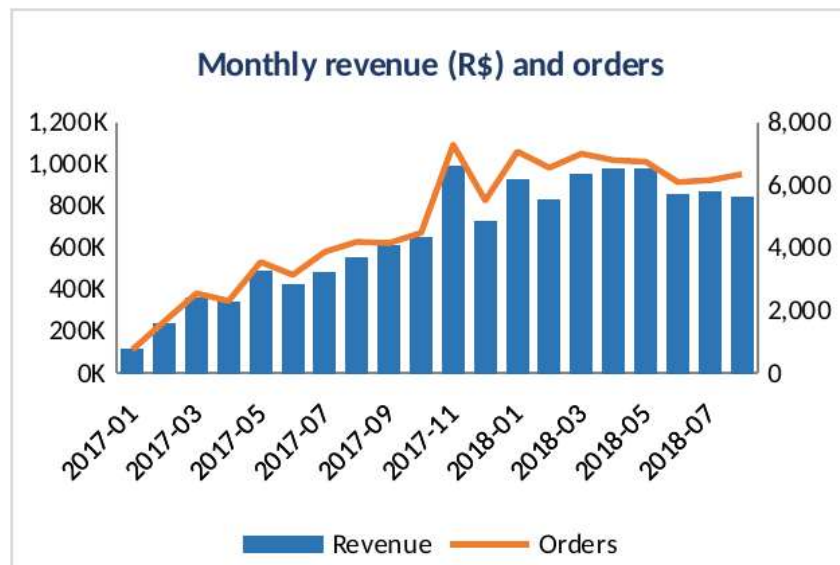
8. Business Questions and Findings

Business rules. Revenue is the sum of item prices in delivered orders (freight excluded). An order is *late* if it was delivered on a date after the estimated delivery date. Customers are counted by `customer_unique_id`. Each answer below comes from one of the 20 SQL queries (Q1 to Q20) described in Section 10.

#	Business question	Finding
1	How big is the business?	96,478 delivered orders, 93,358 customers, R\$ 13.22M revenue, AOV R\$ 137.04
2	How many orders fail to complete?	97.02% are delivered. 0.63% are cancelled and 0.61% are unavailable
3	How are sales trending?	Revenue grew from R\$ 112K (Jan 2017) to a R\$ 988K peak in Nov 2017. In 2018 it stayed between roughly R\$ 0.83M and 0.98M per month. Jan to Aug 2018 revenue was R\$ 7.2M against R\$ 3.0M a year earlier (+141%)
4	How did each year perform?	2017: 43,428 orders and R\$ 5.96M. 2018 (to August): 52,783 orders and R\$ 7.22M. AOV stayed near R\$ 137
5	Which weekday is busiest?	Monday (15,701 orders). Saturday is lowest (10,555)
6	When do customers buy?	Activity runs from 10:00 to 21:00 with peaks near 11:00 and 16:00. The 3:00 to 5:00 period is almost empty
7	Which categories drive revenue?	health_beauty (R\$ 1.23M, 9.3%), watches_gifts (8.8%), bed_bath_table (7.7%). The top 3 hold 25.9%
8	Is revenue concentrated (80/20)?	Yes. 18 of 74 categories (24%) generate about 81% of revenue
9	Which states matter and where is delivery weak?	São Paulo (SP) is 38% of revenue (R\$ 5.07M). Late rates are highest in MA (17.4%), CE (13.8%), BA (12.2%) and RJ (12.1%), against 4.5% in SP. Average delivery takes 8.7 days in SP but 19 to 21 days in BA, CE and MA
10	How do customers pay?	Credit card is 78.5% of payment value, boleto 18.0%, voucher 2.2%, debit 1.4%
11	Do customers use instalments?	Yes, mostly for larger tickets. The average payment is R\$ 95.60 when paid in full and R\$ 330.87 for 7 to 10 instalments
12	Is delivery improving?	Average delivery fell to 7.7 days by Aug 2018. But the late rate spiked to 12.4% (Nov 2017), 14.1% (Feb 2018) and 19.0% (Mar 2018), against roughly 2% to 7% in normal months
13	Do late orders get worse reviews?	Yes. 2.27 stars (late) against 4.29 (on time). Negative reviews (1 to 2 stars): 62.4% against 9.2%
14	How fast does satisfaction drop with delay?	4.32 (very early), 4.04 (on the date), 3.29 (1 to 3 days late), 2.11 (4 to 7 days), 1.70 (8+ days late)
15	Who are the top sellers?	The top seller earned R\$ 227K (1.7% of revenue). 8 of the top 10 are in SP. One top-10 seller has a 3.35 review score despite R\$ 187K revenue
16	Which sellers have the worst experience?	10 sellers with 30+ orders score 2.27 to 3.08 and are late on up to 32% of items. Together they hold 855 orders and R\$ 113K
17	What share of customers buy again?	3.00% (2,801 of 93,358)
18	How many orders do customers place?	97.0% place one order, 2.8% two, 0.2% three, 0.05% four or more. Repeat customers produce 5.5% of revenue
19	Do cohorts come back?	Average month-1 retention is below 1% for every cohort (about 0.5%)
20	Which customer segments matter?	Champions: 16.5% of customers, 31.5% of revenue. At-risk big spenders: 15.6% of customers, 30.6% of revenue, about 394 days since last purchase



Average review score by delivery delay (Excel chart)



Monthly revenue and orders (Excel chart)

9. SQL Analysis

All data preparation and analysis was done in **MySQL 8** using four scripts in the `sql/` folder.

9.1 Database design

Eight tables were created in a database named `olist_db`.

Design choice	Reason
Primary keys on every table (composite keys for order items, payments and reviews)	Guarantees one row per real record
CHAR(32) for IDs	The IDs are fixed-length hashes
CHAR(5) for zip prefixes	Keeps leading zeros, such as 09790
DECIMAL(10,2) for prices, freight and payments	Exact money values with no rounding drift
DATETIME for all timestamps	Allows date arithmetic such as DATEDIFF
Indexes on join and filter columns (customer, product, seller, status, purchase date)	Faster joins on 100,000+ row tables

9.2 Loading the data

Each file was loaded with `LOAD DATA LOCAL INFILE`. Empty cells in date and number columns were converted to NULL with `NULLIF`. A closing query compares the loaded row count of every table against the expected count, and all 8 tables returned OK:

Table	Rows loaded
customers	99,441
sellers	3,095
category_translation	73
products	32,951
orders	99,441
order_items	112,650
order_payments	103,886
order_reviews	99,224

9.3 Data quality checks

#	Check	Result
A1	Date range of orders	4 Sep 2016 to 17 Oct 2018
A2	NULL dates in orders	160 approval, 1,783 carrier, 2,965 delivery
A3	Delivered orders without a delivery date	8 (excluded from delivery analysis)
A4	Orders without items	775
A5	Orphan product, seller and order references	0, 0, 0
A6	Products without a category	610 (labelled "unknown")
A7	Orders with more than one review	547 (averaged per order)
A8	Orders with more than one payment	2,961 (summed per order)
A9	Orders delivered before they were purchased	0
A10	Payment types present	credit_card, boleto, voucher, debit_card and 3 rows of not_defined

9.4 Analysis tables

Two tables were built to make the analysis simple and safe.

- **order_summary**: one row per order, with delivery days, delay days, a late flag, revenue, freight, payments and average review.
- **item_detail**: one row per item sold, with English category, seller state and the order's delivery and review fields.

Avoiding double counting. An order can have several items, several payments and several reviews. Joining these tables directly would repeat rows and inflate revenue. Items, payments and reviews were therefore *aggregated to one row per order first* and joined afterwards. A control query confirms the result: revenue in the raw items table and in `item_detail` is identical at **R\$ 13,591,643.70**, with 99,441 rows in `order_summary` and 112,650 in `item_detail`.

9.5 SQL techniques used

Technique	Where used
Multi-table joins (JOIN, LEFT JOIN)	Building <code>order_summary</code> and <code>item_detail</code>
Aggregation (SUM, COUNT DISTINCT, AVG)	KPI, state, category, seller queries
Common table expressions (WITH)	Monthly trend, Pareto, repeat customers, cohorts, RFM
LAG()	Month-over-month revenue growth (Q3)
RANK()	Category revenue ranking (Q7)
Running totals (SUM() OVER)	Cumulative revenue (Q3), Pareto cumulative share (Q8)
FIRST_VALUE()	Cohort retention percentage (Q19)
NTILE(5)	Recency and spend scores for RFM (Q20)
CASE WHEN	Delay buckets, instalment groups, ABC class, RFM segments
Date functions (DATEDIFF, DATE_FORMAT, HOUR, DAYNAME)	Delivery time, lateness, time patterns
HAVING and LIMIT	Seller filters (30+ orders), top 10 lists

9.6 Query catalogue

Q	Excel sheet	Business question	Main technique
1	01_KPI_Summary	Headline KPIs	Aggregation
2	02_Order_Status	Order status mix	Window total
3	03_Monthly_Trend	Monthly orders, revenue, growth	CTE, LAG, running total
4	04_Yearly_Summary	Yearly performance	Aggregation
5	05_Weekday_Pattern	Orders by weekday	Group by
6	06_Hourly_Pattern	Orders by hour	Group by
7	07_Category_Performance	Revenue by category	RANK, share of total
8	08_Pareto_ABC	80/20 rule and ABC class	Cumulative window, CASE
9	09_State_Performance	Revenue and delivery by state	Multi-metric aggregation
10	10_Payment_Types	Payment method mix	Join, window share
11	11_Credit_Installments	Instalment groups	CASE
12	12_Delivery_Trend	Delivery time and late % by month	Aggregation
13	13_Late_vs_OnTime	Reviews for late vs on-time orders	CASE

Q	Excel sheet	Business question	Main technique
14	14_Delay_Buckets	Review score by delay bucket	Subquery, CASE
15	15_Top_Sellers	Top 10 sellers	Aggregation, LIMIT
16	16_Risky_Sellers	Lowest-rated sellers (30+ orders)	HAVING
17	17_Repeat_Customers	Repeat customer rate	CTE
18	18_Order_Frequency	Customers by number of orders	CTE, CASE
19	19_Cohort_Retention	Retention by first-purchase month	CTE, FIRST_VALUE
20	20_RFM_Segments	Customer segments	CTE, NTILE

10. Dashboard Development

Tool: Microsoft Excel.

10.1 Workbook structure

Sheet	Purpose
One_Page_Dashboard	Executive dashboard: 6 KPI cards, 3 insights, 6 charts on one screen
Full_Dashboard	Longer version with 10 charts
Cohort_Heatmap	Retention grid by first-purchase month, with colour scale
About	Business rules and definitions
01 to 20	One sheet per SQL result, formatted as Excel tables

10.2 Design choices

- **Audience:** executives who need the story on one screen.
- **Layout:** KPI cards, three key insights, then six charts in two rows of three.
- **Colour:** navy and blue for normal values, orange to highlight problems (late deliveries, poor reviews).
- **Date range:** monthly charts use January 2017 to August 2018, because other months have very few orders.

10.3 Dashboard contents

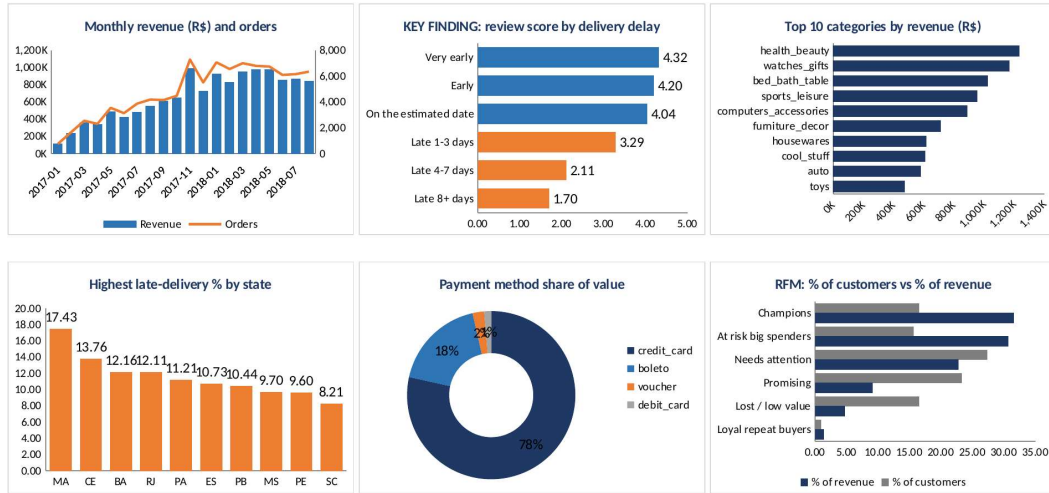
Element	Source sheet	Purpose
Total revenue, orders, AOV, delivery time, late %, review score	01_KPI_Summary	Headline numbers
Monthly revenue and orders	03_Monthly_Trend	Growth and plateau
Review score by delivery delay	14_Delay_Buckets	The central finding
Top 10 categories	07_Category_Performance	Where revenue comes from
Highest late % by state	09_State_Performance	Where delivery is weakest
Payment method share	10_Payment_Types	Payment mix
RFM segments	20_RFM_Segments	Customers by value

OLIST E-COMMERCE DASHBOARD

Sep 2016 - Oct 2018 | delivered orders | amounts in Brazilian Real (R\$)

TOTAL REVENUE	DELIVERED ORDERS	AVG ORDER VALUE	AVG DELIVERY TIME	LATE DELIVERY RATE	AVG REVIEW SCORE
R\$ 13.2M	96,478	R\$ 137.04	12.5 days	6.77%	4.16 / 5
Item sales, excluding freight	93,358 unique customers	Per delivered order	Estimated: 24.4 days	Delivered after estimated date	Average per order

- DELIVERY HURTS REVIEWS: late orders average 2.27 stars vs 4.29 for on-time orders; 62% of late orders get a negative review (1-2 stars) vs 9%.
- CONCENTRATION: Sao Paulo (SP) generates 38% of revenue; the top 3 categories make up 25.9% of sales.
- RETENTION GAP: only 3.0% of customers ever bought a second time, so growth depends on acquiring new customers.



One-page Excel dashboard

Customer retention by first-purchase month (cohort)

Each row is a cohort of customers who first bought in that month. Columns show the % of that cohort who bought again N months later (month 0 = first purchase = 100%). Darker = higher retention.

		Months since first purchase												
Cohort	Cohort size	0	1	2	3	4	5	6	7	8	9	10	11	12
2017-01	717	100.00%	0.28%	0.28%	0.14%	0.42%	0.14%	0.42%	0.14%	0.14%		0.42%	0.14%	0.70%
2017-02	1,628	100.00%	0.18%	0.31%	0.12%	0.43%	0.12%	0.25%	0.18%	0.12%	0.18%	0.12%	0.31%	0.12%
2017-03	2,503	100.00%	0.44%	0.36%	0.40%	0.36%	0.16%	0.16%	0.32%	0.32%	0.08%	0.36%	0.12%	0.20%
2017-04	2,256	100.00%	0.62%	0.22%	0.18%	0.27%	0.27%	0.35%	0.31%	0.31%	0.18%	0.27%	0.09%	0.04%
2017-05	3,451	100.00%	0.46%	0.46%	0.29%	0.29%	0.32%	0.41%	0.14%	0.26%	0.26%	0.26%	0.35%	0.23%
2017-06	3,037	100.00%	0.49%	0.40%	0.43%	0.30%	0.40%	0.36%	0.23%	0.13%	0.20%	0.30%	0.36%	0.16%
2017-07	3,752	100.00%	0.53%	0.35%	0.24%	0.29%	0.21%	0.32%	0.11%	0.19%	0.27%	0.21%	0.29%	0.13%
2017-08	4,057	100.00%	0.69%	0.35%	0.27%	0.35%	0.52%	0.30%	0.27%	0.15%	0.15%	0.25%	0.20%	0.12%
2017-09	4,004	100.00%	0.70%	0.55%	0.27%	0.45%	0.22%	0.22%	0.25%	0.27%	0.17%	0.25%	0.07%	
2017-10	4,328	100.00%	0.72%	0.25%	0.09%	0.23%	0.21%	0.21%	0.37%	0.28%	0.18%	0.21%		
2017-11	7,060	100.00%	0.57%	0.37%	0.17%	0.17%	0.18%	0.11%	0.18%	0.13%	0.06%			
2017-12	5,338	100.00%	0.21%	0.28%	0.34%	0.26%	0.21%	0.17%	0.02%	0.19%				
2018-01	6,842	100.00%	0.34%	0.37%	0.29%	0.29%	0.16%	0.18%	0.23%					
2018-02	6,288	100.00%	0.35%	0.40%	0.30%	0.25%	0.22%	0.21%						
2018-03	6,774	100.00%	0.40%	0.30%	0.30%	0.12%	0.12%							
2018-04	6,582	100.00%	0.59%	0.30%	0.24%	0.14%								
2018-05	6,506	100.00%	0.52%	0.26%	0.18%									
2018-06	5,878	100.00%	0.43%	0.27%										
2018-07	5,949	100.00%	0.52%											
2018-08	6,144	100.00%												

Reading the chart: retention after the first month is below 1% for almost every cohort, so almost no customers come back. This is the single biggest growth opportunity.

Cohort retention heat map (Excel)

10.4 Validation: dashboard against SQL results

Measure	SQL result (Q1)	Dashboard card
Total revenue	13,221,498.11	R\$ 13.2M
Delivered orders	96,478	96,478
Average order value	137.04	R\$ 137.04
Average delivery time	12.5	12.5 days
Late delivery rate	6.77	6.77%
Average review score	4.16	4.16 / 5
Review score for 8+ days late (Q14)	1.70	1.70

The cards and charts read straight from the query-result sheets, so the dashboard and the SQL output cannot disagree.

11. Business Recommendations

Rank	Recommendation	Evidence	Owner	KPI to track
1	Stop the worst delays. Monitor open orders as they approach the promised date, alert customers early, and escalate any order more than 3 days overdue	Orders 8+ days late: 2,862, with 79% negative reviews and a 1.70 score	Logistics	Orders 4+ days late; late %
2	Prepare for peaks. Agree extra carrier capacity before November and run daily late-order reviews in peak months	Late rate hit 12.4% (Nov 2017), 14.1% (Feb 2018) and 19.0% (Mar 2018)	Logistics, Seller management	Late % in peak months
3	Fix regional delivery. Add regional carriers or hubs for MA, CE, BA, AL, SE and RJ, and recruit sellers closer to these regions	Late rates of 12% to 21% against 4.5% in SP. Delivery takes 8.7 days in SP but 19 to 25 days in these northern states. 8 of the top 10 sellers are in SP	Logistics, Marketplace	Late % and delivery days by state
4	Run seller scorecards. Track review score and late % per seller. Warn and coach the weakest, and set minimum standards for high-revenue sellers	10 sellers score 2.27 to 3.08. A top-10 seller scores 3.35	Seller management	Seller review score; late item %
5	Build a retention programme. Send a second-purchase offer in the first 30 to 60 days and use CRM follow-up	Only 3.0% repeat. Month-1 retention is about 0.5%	Marketing / CRM	Repeat rate; month-1 retention
6	Win back at-risk big spenders with a targeted offer	14,563 customers (15.6%) hold 30.6% of revenue and have been inactive about 394 days	Marketing / CRM	Reactivation rate
7	Review the delivery promise. Test tighter, route-based estimates on fast routes while protecting the late tail	Estimate 24.4 days against 12.5 days actual. 71,303 orders (74%) arrive 8+ days early	Logistics, Product	Late %; conversion in a test
8	Review freight in distant regions. Test free-shipping thresholds or subsidies for high-freight states	Freight is 13.9% of revenue in SP but 26% in MA	Pricing, Finance	Freight %; conversion
9	Protect core categories. Focus on the 18 A-class categories and fix quality in bed_bath_table (3.92 score, #3 by revenue)	24% of categories give about 81% of revenue	Merchandising	Category review score

- **Quick wins (0 to 30 days):** late-order alerts, seller watch-list, retention email trial.
- **Medium term (1 to 3 months):** peak plan, seller scorecard, win-back campaign.
- **Longer term (3 to 12 months):** regional logistics, delivery promise redesign, freight tests.

12. Business Impact

These are **scenario estimates** calculated from the SQL results to size the opportunity. They are not forecasts and assume the relationships in the data would hold.

Scenario	Assumption	Estimated effect
Cut late orders by 25%	1,634 of 6,534 late orders become on-time, taking the on-time review average (4.29 instead of 2.27)	Average review rises from 4.16 to about 4.19
Cut late orders by 50%	3,267 late orders become on-time	Average review rises to about 4.23, with about 1,700 fewer 1 to 2 star reviews
Raise repeat rate by 1 point	934 extra repeat customers at the current AOV of R\$ 137.04	About R\$ 128K extra revenue (about 1.0% of the total)
Reactivate 5% of at-risk big spenders	728 customers at their average spend of R\$ 278	About R\$ 202K extra revenue (about 1.5% of the total)
Fix or exit the 10 weakest sellers	Replace their 855 orders with average-quality service	Protects about R\$ 113K revenue (0.9%) and removes a source of 1 to 2 star reviews

Caveats. The review improvement is an upper-bound illustration, because part of the review gap may be caused by other factors such as the product or the seller. Revenue figures exclude freight and profit, because the data has no cost information. Results should be tested with a pilot before budget is committed.

Strategic value. The main benefit is a shift from relying on new-customer growth to protecting satisfaction and repeat business, which lowers acquisition cost over time.

13. Conclusion and Future Scope

13.1 Conclusion

The marketplace is large and growing, with R\$ 13.2 million in revenue from 96,478 delivered orders. The analysis shows that its main weakness is experience and loyalty rather than sales volume:

1. **Delivery drives satisfaction.** Late orders score 2.27 stars against 4.29, and the score keeps falling as delays grow, from 4.32 for very early deliveries to 1.70 for orders 8 or more days late.
2. **Delivery problems are concentrated** in specific regions (the North, Northeast and Rio de Janeiro), specific months (peak periods) and a small group of weak sellers.
3. **Very few customers return.** Only 3.0% buy a second time, so growth depends on constant new-customer acquisition.
4. **Revenue is concentrated** in São Paulo (38%) and in 18 of 74 categories (about 81%).
5. **There is real upside.** About 31% of revenue sits with high-spend customers who have gone quiet, and delivery promises are very cautious (74% of orders arrive 8 or more days early).

Acting first on late-delivery control, regional logistics, seller standards and retention offers the highest expected return.

13.2 Future scope

Area	Idea
Predictive modelling	Predict late delivery and negative reviews, for example in Python, to flag at-risk orders before delivery
Forecasting	Forecast monthly demand and delivery load for peak planning
Text analytics	Analyse the Portuguese review comments to find the complaint themes behind low scores
Geospatial analysis	Use the geolocation file to study seller-to-customer distance, delivery time and freight
Customer lifetime value	Estimate CLV and acquisition payback by segment
Market basket analysis	Find products bought together for bundles
Experiments	A/B test delivery promises, free-shipping thresholds and win-back offers
BI tools	Rebuild the dashboard in Power BI or Tableau and schedule the SQL refresh
Profitability	Add cost and margin data to move from revenue to profit analysis